

Student Performance Analysis

Principal's Report — 500 Students

Dataset scope: 500 students.

Observed pass rate: 97.6% | Observed fail rate: 2.4% (12 students).

Important data-quality note: the prompt states that 30% fail every year, but this 500-student extract shows a much lower observed failure rate. The report uses the uploaded data rather than assuming the 30% figure.

The strongest observed relationship with total score is weekly self-study (correlation 0.80). Attendance and class participation show much weaker linear relationships in this sample.

For early intervention, risk segmentation is based only on attendance, self-study hours and class participation—not final score—so it can be used as an early-warning screen.

Key Findings & Recommendations

1. Self-study is the clearest academic signal in this dataset. Students with lower weekly self-study hours tend to have substantially lower total scores.
2. The early-warning segmentation identifies students who need attention before relying on the final outcome.
3. Moderate-risk students are the largest intervention group; they should receive routine monitoring rather than waiting for failure.
4. High/Critical-risk students should receive an individual support plan covering attendance, study routine and classroom engagement.
5. Recommendations:
 - Create a weekly early-warning dashboard.
 - Contact students showing repeated low attendance.
 - Provide structured study plans and tutoring for low-study students.
 - Encourage active classroom participation through low-stakes activities.
 - Review risk status every 2-4 weeks and record intervention outcomes.
 - Treat risk labels as support signals, not permanent labels or punishment.

Correlation & Factor Analysis

Correlation matrix:

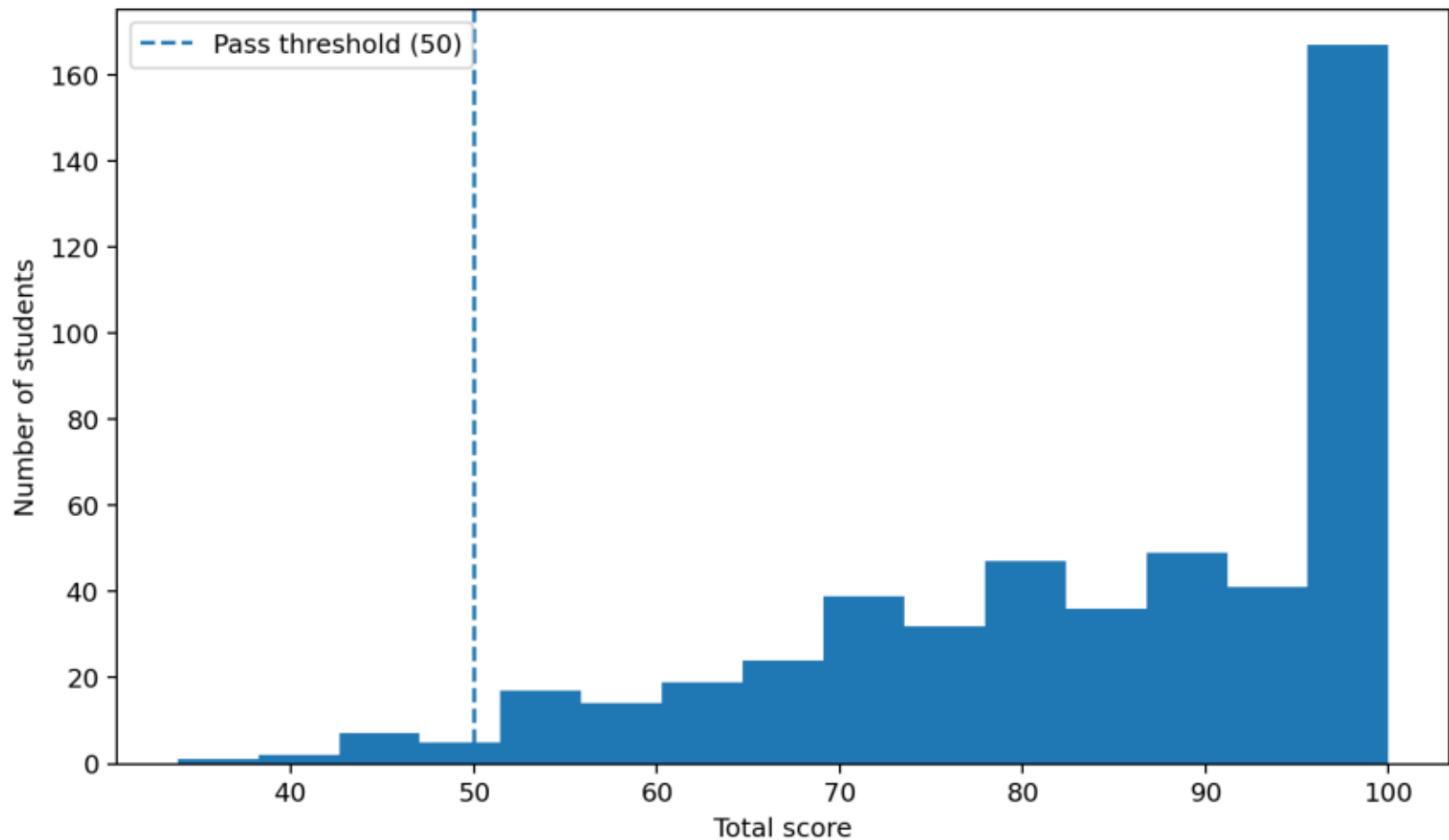
	weekly_self_study_hours	attendance_percentage	class_participation	total_score
weekly_self_study_hours	1.00	-0.02	0.02	0.80
attendance_percentage	-0.02	1.00	0.01	-0.01
class_participation	0.02	0.01	1.00	0.02
total_score	0.80	-0.01	0.02	1.00

Exploratory factor loadings (standardized variables):

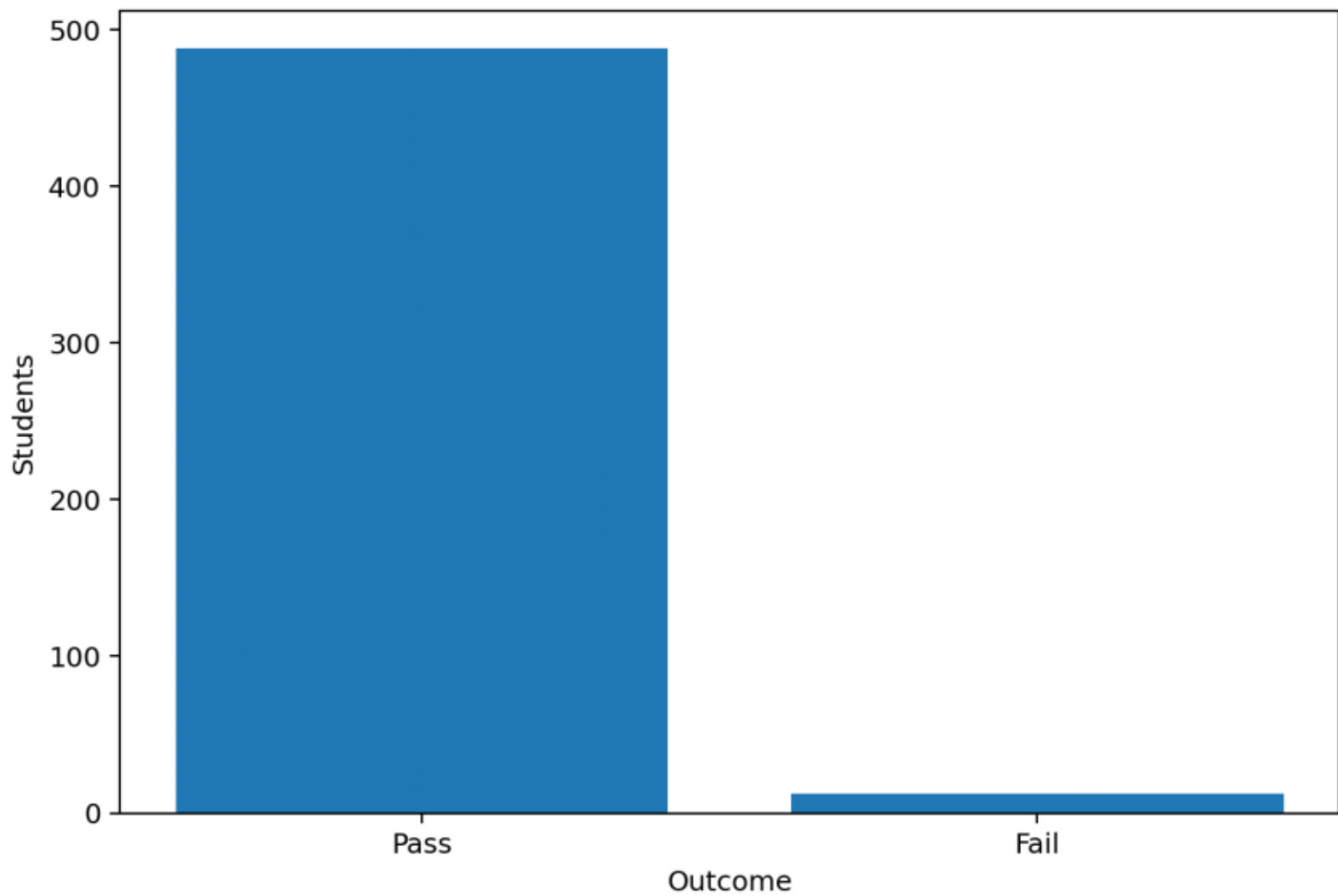
	Factor 1	Factor 2
weekly_self_study_hours	-0.894	0.030
attendance_percentage	0.013	-0.145
class_participation	-0.021	-0.058
total_score	-0.894	-0.030

Interpretation: Factor 1 is dominated by self-study and total score, indicating a strong academic-performance/study dimension. The second factor is comparatively weak in this small variable set. Factor analysis here is exploratory and should not be treated as a definitive causal model.

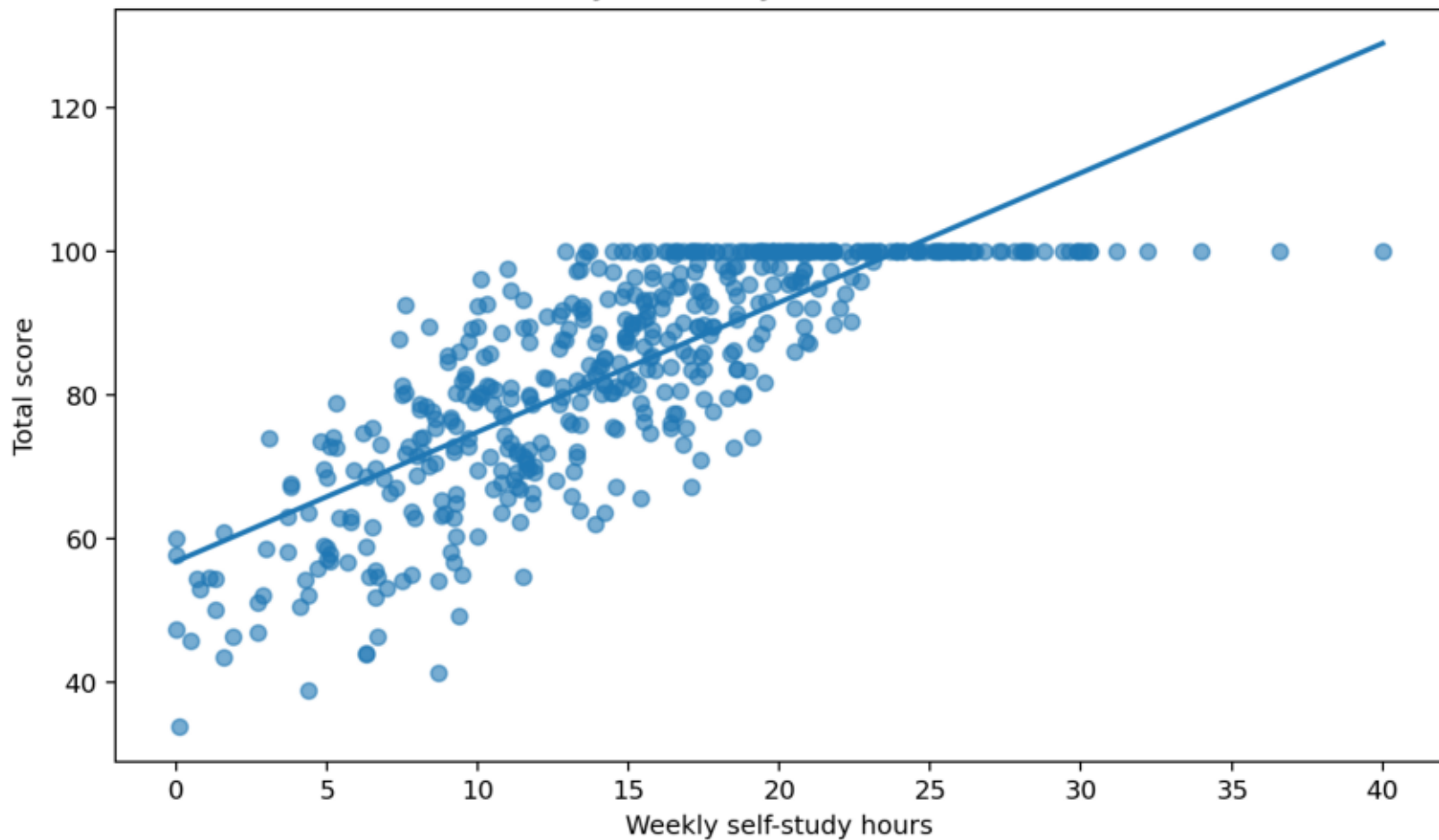
1. Distribution of Total Scores



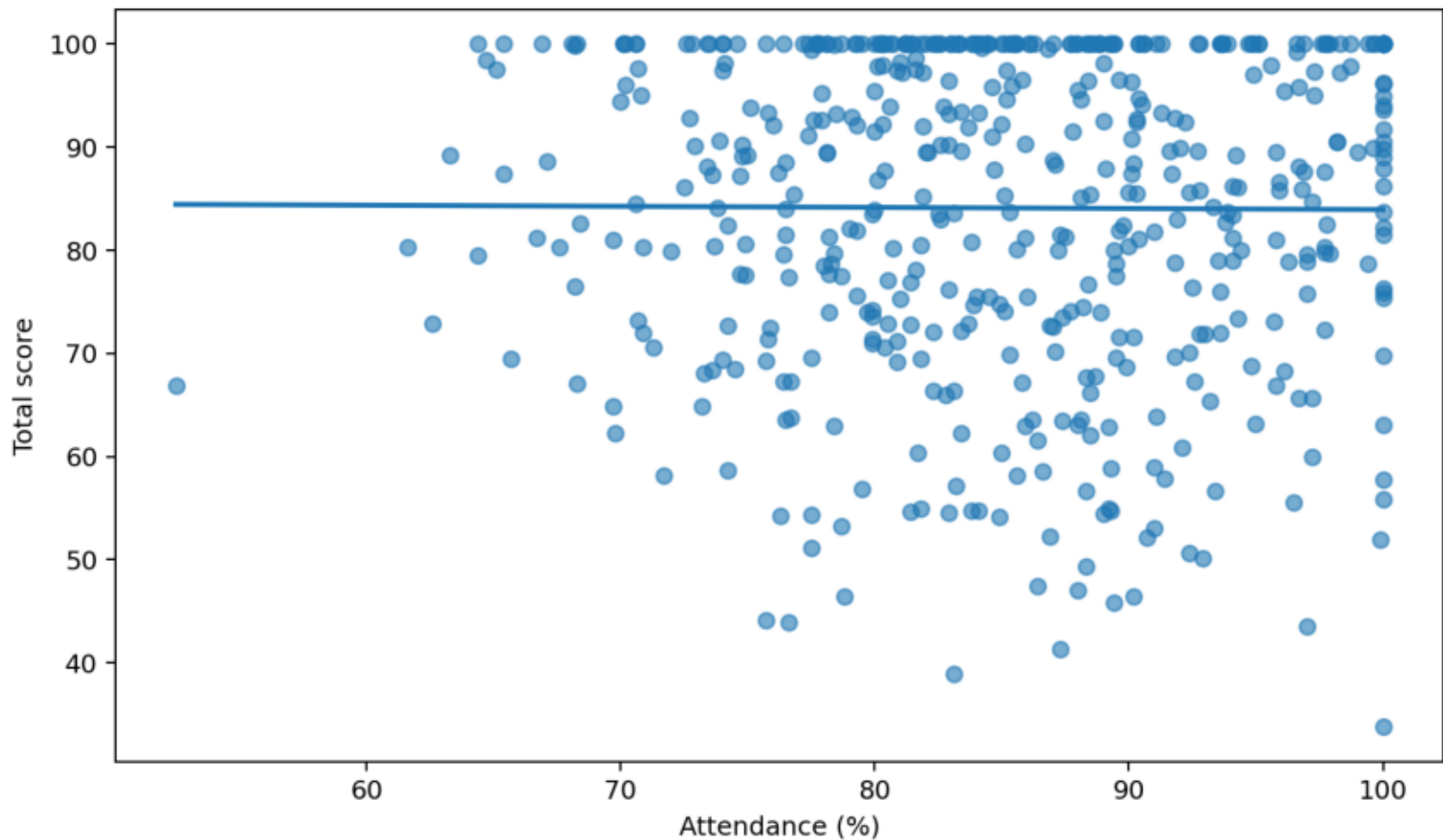
2. Pass vs Fail



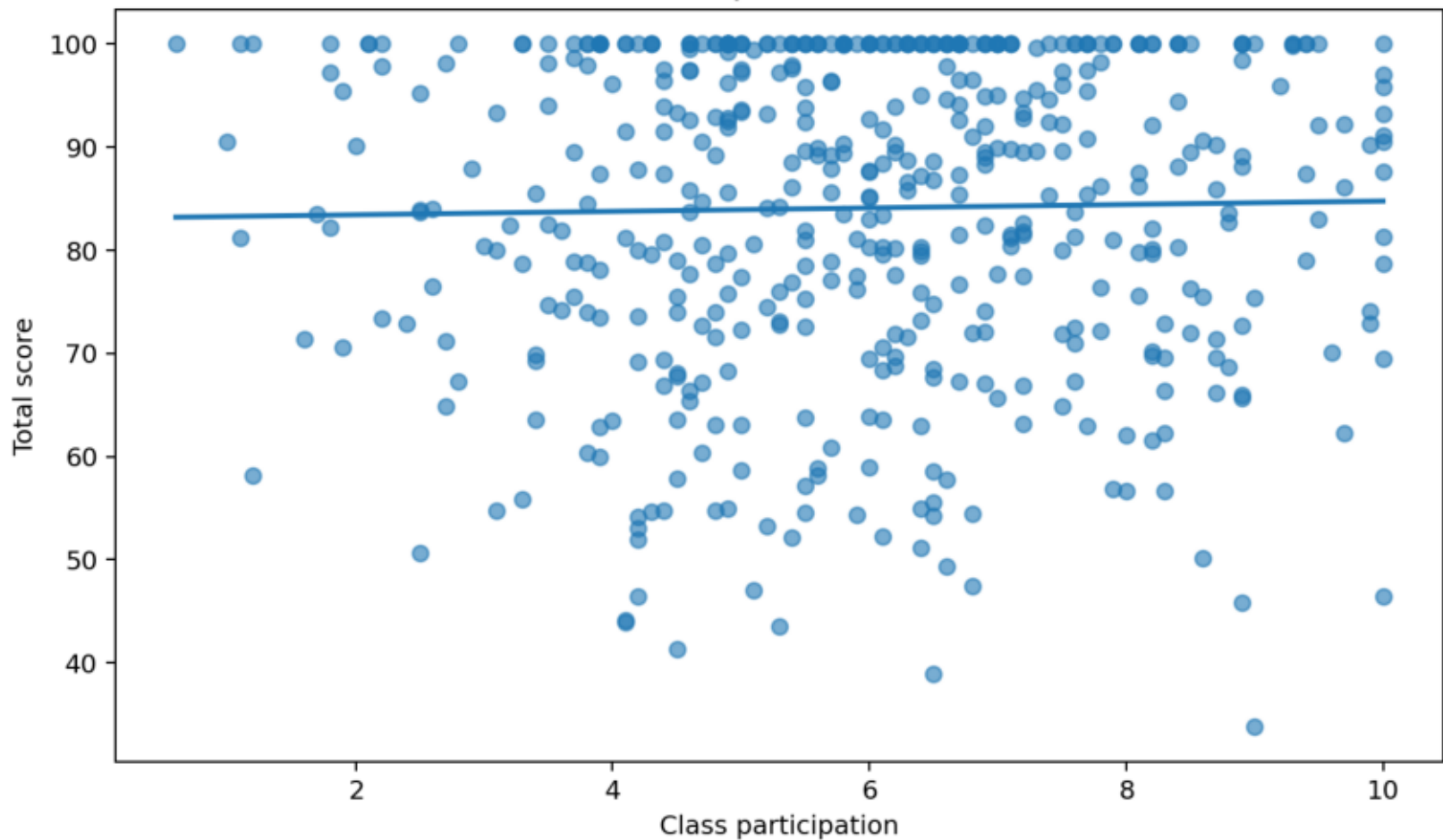
3. Weekly Self-Study Hours vs Total Score



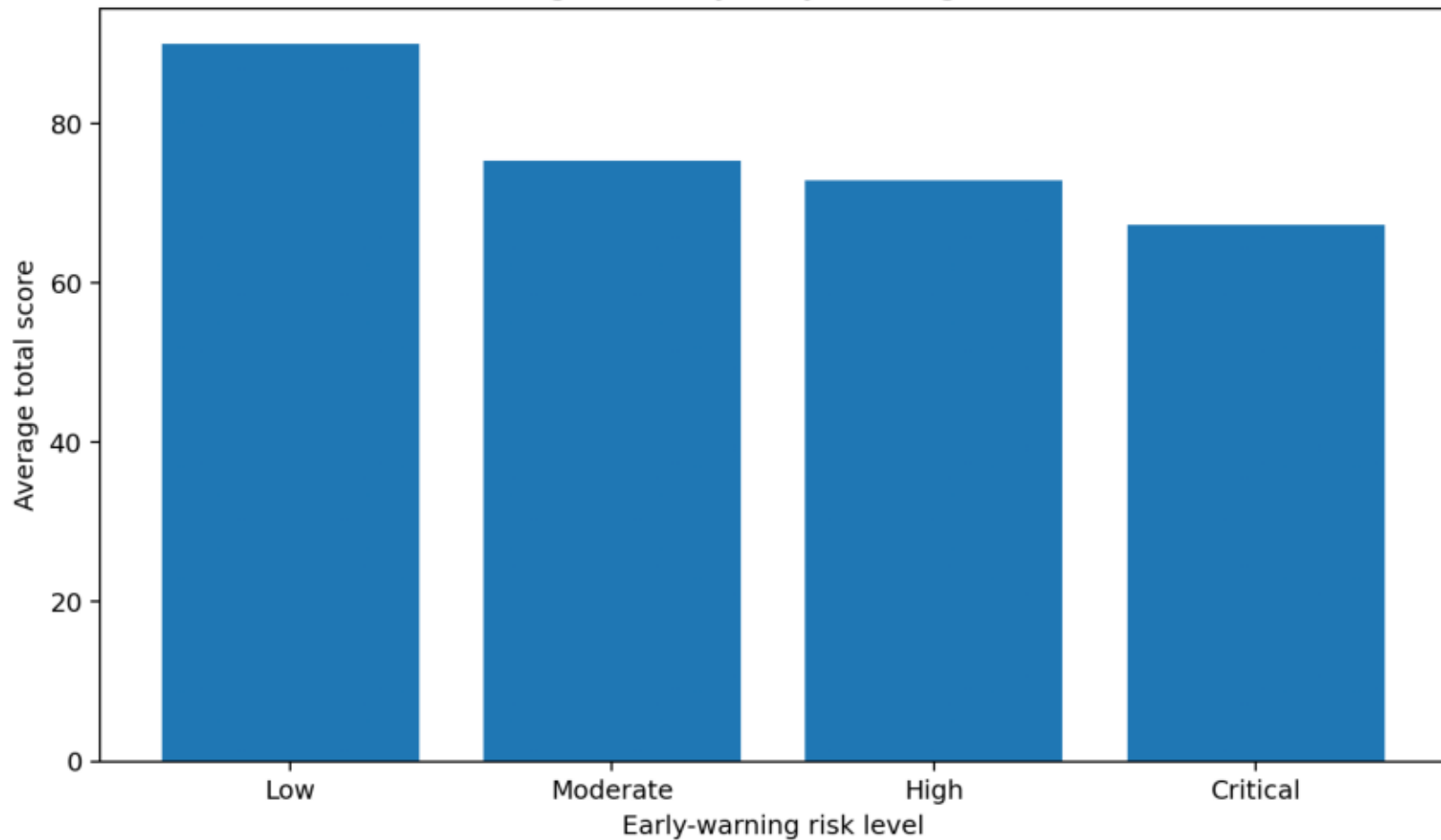
4. Attendance vs Total Score



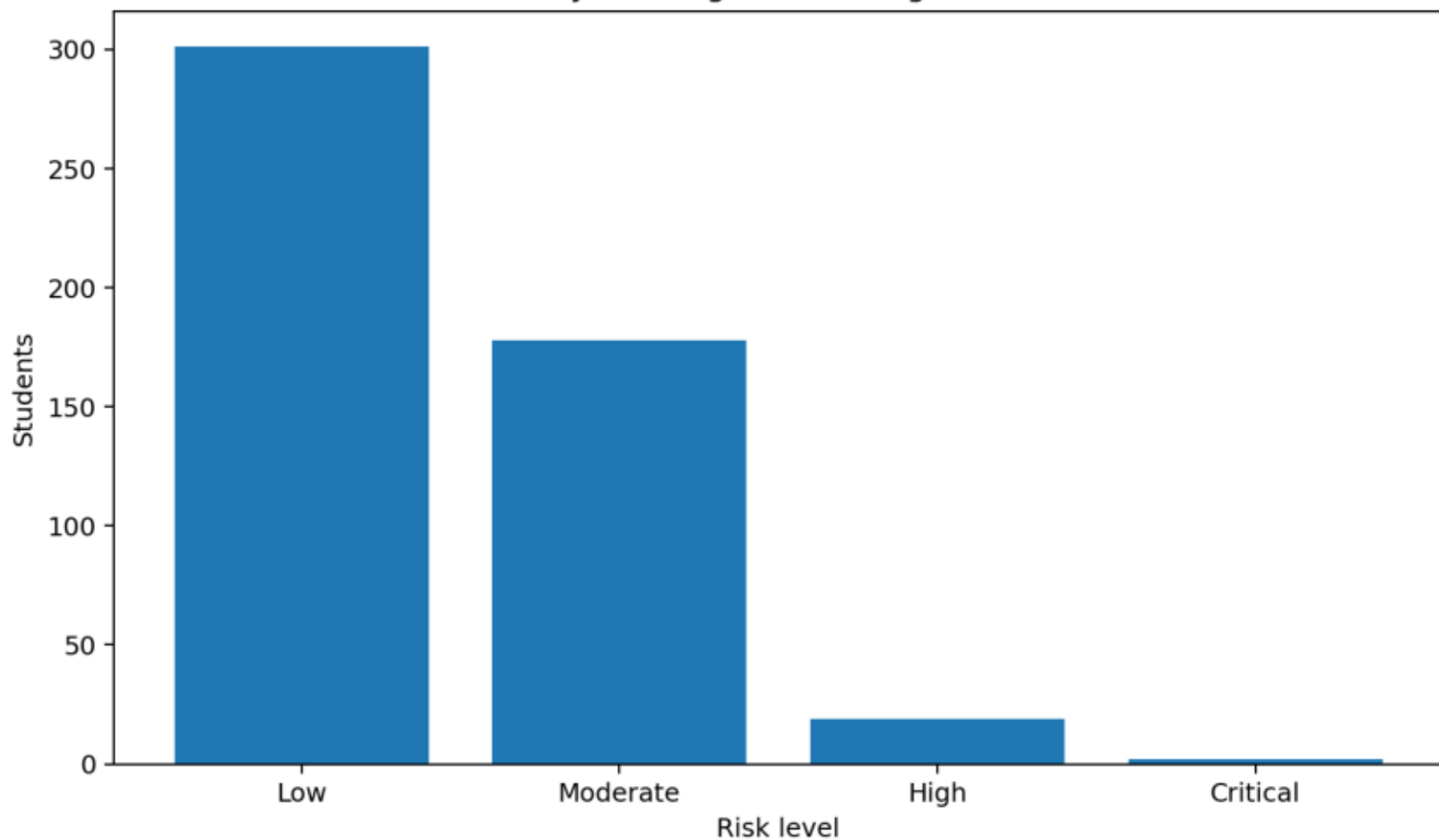
5. Class Participation vs Total Score



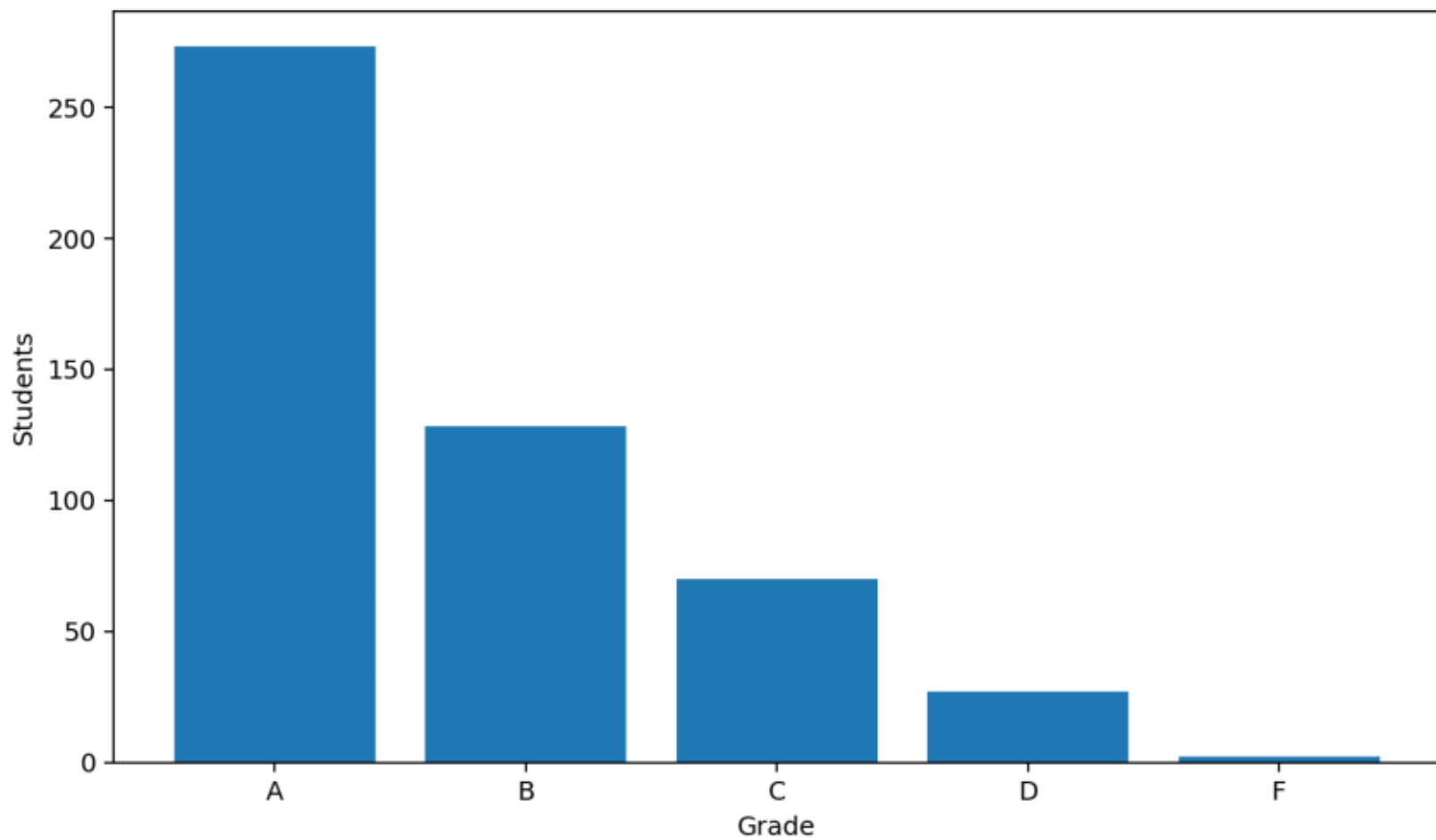
6. Average Score by Early-Warning Risk Level



7. Early-Warning Student Segmentation



8. Grade Distribution



9. Correlation Heatmap

